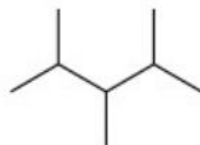


Thermochemistry – Question Set

- 1) (Exercise 5.7 page 251) Gasoline is composed primarily of hydrocarbons, including many with eight carbon atoms, called octanes. One of the cleanest-burning octanes is a compound called 2,3,4-trimethylpentane (C_8H_{18}), which has the following structural formula:



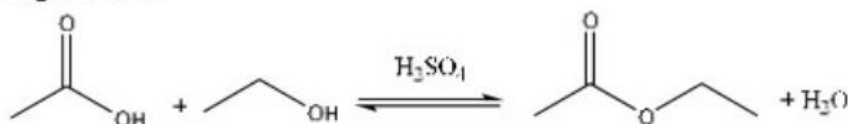
The complete combustion of one mole of this compound to $CO_2(g)$ and $H_2O(g)$ leads to $\Delta H^\circ = -5064.9$ kJ. (a) Write a balanced equation for the combustion of 1 mol of $C_8H_{18}(l)$. (b) By using the information in this problem and data in Table 5.3 in the book (page 231), calculate ΔH_f° for 2,3,4-trimethylpentane.

- 2) Consider the following polymerisation reaction:

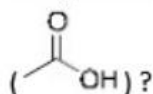


- Explain why ΔH° and ΔS° are negative ($\Delta H^\circ = -109$ kJ mol⁻¹; $\Delta S^\circ = -155$ J mol⁻¹ K⁻¹)
- Calculate ΔG°
- Calculate at which temperature polymerisation stops being spontaneous
- What limitations does the calculation in c) have?

- 3) Consider the following reaction:



- Knowing $\Delta H^\circ = -2$ kJ mol⁻¹ and $\Delta S^\circ = +8$ J mol⁻¹ K⁻¹, calculate the equilibrium constant, K, at 298 K.
- Which reaction will be spontaneous at conversions 0.3, 0.6 and 0.8
- Will your answer be the same if the starting point is twice as much ethanol () as acetic acid



- 4) Predict whether the following reactions will occur with negative, positive or very small (close to 0) entropy change. Explain why:

- $H_2O(l) \rightarrow H_2O(g)$
- $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
- $4 Fe(s) + 3 O_2(g) \rightarrow Fe_2O_3(s)$
- $N_2(g) + O_2(g) \rightarrow 2NO(g)$

- 5) Consider the reaction: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$, $\Delta H^\circ = -802.3$ kJ mol⁻¹; $\Delta S^\circ = -5$ J mol⁻¹ K⁻¹

- Will this reaction be spontaneous at standard conditions?
- Calculate the equilibrium constant, K.
- What temperature will this reaction cease to be spontaneous?
- Discuss the validity of these calculations.